

ASSIGNMENT No 1

```
import java.util.*;

public class GraphTraversal {

    private int vertices;
    private LinkedList<Integer>[] adj;

    @SuppressWarnings("unchecked")
    public GraphTraversal(int v) {
        vertices = v;
        adj = new LinkedList[v];
        for (int i = 0; i < v; i++) {
            adj[i] = new LinkedList<>();
        }
    }

    // Add edge (Undirected)
    public void addEdge(int v, int w) {
        adj[v].add(w);
        adj[w].add(v);
    }

    // ----- BFS -----
    public void BFS(int start) {
        boolean[] visited = new boolean[vertices];
        Queue<Integer> queue = new LinkedList<>();

        visited[start] = true;
        queue.add(start);

        System.out.print("BFS Traversal: ");

        while (!queue.isEmpty()) {
            int vertex = queue.poll();
            System.out.print(vertex + " ");

            for (int neighbour : adj[vertex]) {
                if (!visited[neighbour]) {
                    visited[neighbour] = true;
                    queue.add(neighbour);
                }
            }
        }
        System.out.println();
    }
}
```

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// ----- DFS -----
public void DFS(int start) {
    boolean[] visited = new boolean[vertices];
    System.out.print("DFS Traversal: ");
    DFSUtil(start, visited);
    System.out.println();
}

private void DFSUtil(int vertex, boolean[] visited) {
    visited[vertex] = true;
    System.out.print(vertex + " ");

    for (int neighbour : adj[vertex]) {
        if (!visited[neighbour]) {
            DFSUtil(neighbour, visited);
        }
    }
}

// ----- MAIN -----
public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    try {
        System.out.print("Enter number of vertices: ");
        int v = sc.nextInt();

        if (v <= 0) {
            System.out.println("Number of vertices must be positive.");
            return;
        }

        GraphTraversal graph = new GraphTraversal(v);

        System.out.print("Enter number of edges: ");
        int e = sc.nextInt();

        System.out.println("Enter edges (source destination):");

        for (int i = 0; i < e; i++) {
            int src = sc.nextInt();
            int dest = sc.nextInt();

            if (src >= 0 && src < v && dest >= 0 && dest < v) {
                graph.addEdge(src, dest);
            } else {
                System.out.println("Invalid edge! Try again.");
            }
        }
    }
}

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        i--;
    }
}

System.out.print("Enter starting vertex for traversal: ");
int start = sc.nextInt();

if (start < 0 || start >= v) {
    System.out.println("Invalid starting vertex.");
    return;
}

System.out.println("\n--- Traversal Results ---");
graph.BFS(start);
graph.DFS(start);

} catch (Exception ex) {
    System.out.println("Invalid input! Please enter integers only.");
} finally {
    sc.close();
}
}
}

```

OUTPUT:-

```

C:\Users\Swaraj\Desktop\Work\LP-2\AI>java GraphTraversal
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
Enter number of vertices: 5
Enter number of edges: 5
Enter edges (source destination):
1 2
2 4
0 3
1 1
2 3
Enter starting vertex for traversal: 2

--- Traversal Results ---
BFS Traversal: 2 1 4 3 0
DFS Traversal: 2 1 4 3 0

```